Problem G
Chess Rating Evaluator

Time limit: 3 seconds
Memory limit: 1024 megabytes

Problem Description

Magnus, a high school student studying computer science, has been drilling chess recently. He plays chess with his friends online quite often. However, he is really dissatisfied with the current Chess rating evaluator that Chess.com, the world biggest online chess platform, is using. So, he has come up with his own rule to evaluate a chess game.

Here are the rules Magnus designed:

1. Each move is given an accuracy rate (0 $\sim$ 100)
2. The moves in different game stages have different weights.
   - The Opening (Moves 1–10): Usually pure memorization. $Weight = 0.8$.
   - The Middlegame (Moves 11–30): The most complicated part. $Weight = 1.2$.
   - The Endgame (Moves 31+): The remaining moves. $Weight = 1.0$.
     Note: You should multiply the accuracy rate by the weight to get the "Weighted Score" of a move.
3. The final score is determine with the following formula:
   - Performance Value $=$ Sum of Weighted Score / Number of Moves ($N - 1$)
   - Final Rating $=$ (Performance Value $\times$ 25) $-$ 500
     Note: The minimum rating is 400. If the calculated result is less than 400, consider it as 400.
4. To award a brilliant move, if the accuracy of a move is 100, Magnus wants you to consider it as 125 for the Weighted Score calculation. To punish a stupid move, if the accuracy of a move is strictly less than 40, Magnus wants you to consider it as 0 for the Weighted Score calculation. For example, if a player made a move that gets a 100 accuracy rate at the sixteenth move, the Weighted Score of that move is $125 \times 1.2 = 150$. If a player made a move that get a 39 accuracy rate at the second move, the Weighted Score of that move is $0 \times 0.8 = 0$.
5. Magnus doesn't want a single mistake to affect the final rating too severely, so before calculating the Performance Value, he wants to remove the move that has the least Weighted Score. Thus, if a game of only six moves has the following Weighted Scores: $[72, 72, 56, 32, 64, 48]$, the Performance Value is $(72 + 72 + 56 + 64 + 48)/5 = 62.4$. The Final Rating of this game is $(62.4 \times 25) - 500 = 1060$.

Your task is to help Magnus implement this Chess Rating Evaluator !

Input Format

The first line of input is an integer T, representing the number of test cases. Each test case contains two lines. The first line is an integer N, representing the number of moves of a game. The second line has N integers, $AR_1, AR_2, \cdots, AR_N$, representing the accuracy rate of each move in the current game.

Output Format

For each case, output a single integer R, which represents the final rating of a game. If R is less than 400, output 400. If the R is a floating-point number, just truncate(floor) any decimal part.

Technical Specification

- $1 \leq T \leq 1,000$
- $2 \leq N \leq 1,000$
- $0 \leq AR_i \leq 100, 1 \leq i \leq N$
- Please use double to perform all the floating-point numbers, do not use float!

Sample Input 1

```
2
6
90 90 70 40 80 60
11
90 90 90 90 90 90 90 90 90 90 100
```

Sample Output 1

```
1060
1495
```

Sample Input 2

```
1
4
65 85 48 73
```

Sample Output 2

```
986
```